

Algorithmic Statistical Arbitrage: Walk-Forward Cointegration & Pairs Trading System

Self-directed work combining statistical time-series methods and Python implementation

PROJECT OVERVIEW:

Exploiting long term statistical equilibrium between non-stationary financial time series (I(1)) by modeling mean-reverting stationary spread residuals (I(0)). The model dynamically trades JPMorgan Chase (JPM) and Wells Fargo (WFC) while enforcing strict causal estimation to eliminate look-ahead bias.

METHODOLOGY

- Stationarity & ADF Testing: Verified non-stationarity of raw price series (I(1)) via Augmented Dickey-Fuller (ADF) tests and confirmed stationarity (I(0)) of linear spread residuals  $\varepsilon_T$
- Causal Walk-Forward Estimation: Applied a 252-day rolling OLS window ( $\beta_T, \alpha_T$ ) with a t-1 lag shift (.shift(1)), eliminating full-sample look-ahead leakage.
- Dynamic Z-Score Normalization: Computed time-varying residuals  $\varepsilon_T = Y_T - (\alpha_{T-1} + \beta_{T-1}X_T)$  and normalized using a 20-day rolling window:  $Z_T = (\varepsilon_T - \mu_{20})/\sigma_{20}$
- Market-Neutral Execution Logic: Triggered entry  $Z_T > 2.0$  (Long/Short spread) and exit at mean-reversion ( $Z_T$  to 0.0) using dynamic leg sizing (1:  $-\beta_T$ ).
- Honest Sample Loss Accounting: Incorporates a 271-day initial burn-in period (252-day rolling  $\beta$  window + 19-day Z-score runway) to guarantee strictly out-of-sample signals.

KEY METRIC & PERFORMANCE

| METRIC                       | VALUE                    |
|------------------------------|--------------------------|
| Annualized Sharpe Ratio      | 1.3648                   |
| Maximum Drawdown             | -\$27.51                 |
| Drawdown Period              | 2024-09-27 to 2025-01-03 |
| Estimated Half Life          | 21.65 Trading days       |
| Honest Sample loss (Burn-In) | 271 Trading Days         |

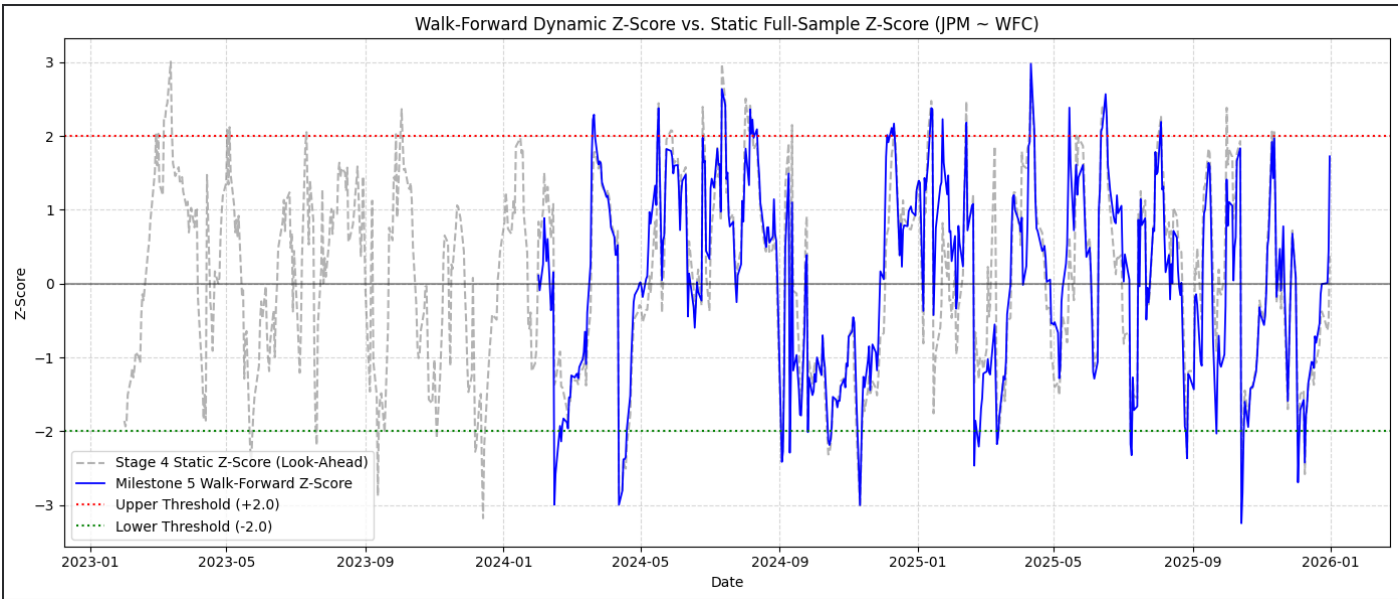


Figure 1 Dynamic vs Static Beta based Z-score